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Summary

I am a Ph.D. candidate with industrial experience in **search** and **recommendation** systems from previous roles at Tencent and Alibaba. My academic work focuses on **representation learning**, **feature modeling**, and **data-centric methods**, which translate well to retrieval, ranking, and LLM-augmented systems. Combining research and large-scale industry experience, I aim to apply machine learning to improve retrieval, ranking, and AI-driven product performance, including systems that integrate LLMs.

Education

Ph.D., Computer Science, Arizona State University, 2023 - 2026
B.E. / M.S., Computer Science, Sichuan University, 2012 - 2019

Professional Experience

1. Full Time, Platform and Content Group

Tencent, Beijing, November 2020 - August 2022

- **Time-Sensitive Query Understanding and Ranking Control.** Designed and deployed a query time-sensitivity detection module for large-scale search systems, improving freshness-aware ranking and user experience in time-sensitive scenarios such as news.
 - Trained a TinyBERT-based semantic model to identify temporal intent from query text and output a time-sensitivity signal.
 - Integrated the time-sensitivity signal into ranking strategies to dynamically balance relevance and freshness for search results.
 - Triggered content aggregation cards at the product level for strongly time-sensitive queries, enhancing user interaction.
- **Event Aggregation and Burst Detection for News Search.** Enhanced news search by detecting, aggregating, and modeling events from retrieved content when query signals were insufficient.
 - Developed title-level event extraction models to identify event semantics and detect bursty event signals from retrieved news candidates.
 - Aggregated news articles into event-level clusters to support event-aware retrieval, de-duplication, and aggregation-based presentation.
 - Built and released a benchmark dataset for event extraction and evaluation (EMNLP), supporting systematic improvement of event modeling in search.
- **Retrieval System Integration and Reliability Improvement.** Improved the robustness and service quality of production search systems through integration with a new content retrieval platform.
 - Collaborated with engineering teams to onboard new content streams and integrate them into the production retrieval pipeline.
 - Validated retrieval coverage and system behavior under real traffic, contributing to reduced failure cases and improved service stability.

2. Full Time, Digital Media & Entertainment Group

Alibaba, Beijing, June 2019 - October 2020

- **Tagging System Implementation.** Built a recommendation-oriented hierarchical tagging system for short-form videos, addressing cold-start and retrieval coverage issues in recommendation pipelines.

- Defined a recommendation-friendly tag taxonomy (1100+ tags) with operations teams.
- Constructed a labeled dataset of 400k+ videos and trained multimodal tagging models.
- Improved tag accuracy and coverage, strengthening content representation quality.
- **Online Impact & Validation.** Exposed the value of the tagging system through online distribution.
 - Integrated tags into retrieval pipelines via tag-based matching and embedding-based vector retrieval.
 - Validated system value through improvements in CTR, watch time, and retention.

3. **Internship**, Data Science & System Security Department NEC Laboratories America, Inc., Princeton, May 2025 - August 2025

- Developed multi-agent LLM frameworks for structured knowledge extraction, producing procedural graph representations to support retrieval-augmented generation (RAG) pipelines.
- Designed structured representations that improve grounding and retrieval consistency, enabling more reliable downstream reasoning and generation.
- Explored how user-specific workflows can be mapped into structured knowledge, informing personalization strategies in LLM-based retrieval systems.

4. **Internship**, Institute of High Performance Computing A*STAR, Singapore, June 2024 - August 2024

- Studied reliability and trustworthiness of LLM-based systems in medical applications, focusing on how adversarial and jailbreak inputs impact system behavior.
- Conducted systematic analysis of failure modes and jailbreak strategies, providing insights applicable to improving robustness and safety in production LLM and retrieval-augmented systems.

Selected Publications

I have published papers in top-tier venues, including KDD, NeurIPS, EMNLP, AAAI, CIKM, IJCAI, and TKDD. A complete list of publications is available on my personal website.

1. **[KDD]** Unsupervised Generative Feature Transformation via Graph Contrastive Pre-training and Multi-objective Fine-tuning.
 - Transforms features into simpler and more discriminative representation spaces, enabling strong downstream performance with lightweight models.
2. **[TKDD]** Feature Selection as Deep Sequential Generative Learning.
 - Formulates feature selection as a sequential generative process, enabling structured and controllable selection of informative features for downstream prediction tasks.
3. **[CIKM]** Revolutionizing Biomarker Discovery: Leveraging Generative AI for Bio-Knowledge-Embedded Continuous Space Exploration.
 - Applied generative feature selection to high-dimensional biological data to identify a compact set of informative features, improving disease prediction performance.
4. **[EMNLP]** Title2Event: Benchmarking Open Event Extraction with a Large-scale Chinese Title Dataset.
 - Event extraction from short news titles, enabling event-level representation, aggregation, and event-driven news search.

Honors

- KDD 2024 Travel Award
- National Scholarship, Top 1%, 2019
- Postgraduate Scholarship in Sichuan University, 1st Prize, Top 1%, 2018, 2019